Venkata Lakshmi Parimala Pasupuleti

Agentic AI Engineer · LangGraph · RAG Pipelines · LLM Evaluation · Generative AI · Production AI · Python · AWS · Cloud-Native

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PROFESSIONAL SUMMARY

Production **Agentic AI + Generative AI** engineer. Built **Leasel**: 8-agent LangGraph pipeline with HITL, RAG, responsible AI guardrails, and 93.6% precision in a regulated enterprise environment - every agent routes through durable state, deterministic compliance checks, and human escalation when confidence is below threshold. Built **AgentProbe**: open-source AI evaluation toolkit with 5-metric eval engine, observability dashboard, and CI/CD regression harness blocking model regressions before production. At **Entrata**: 8 years shipping enterprise production SWE on AWS - 100K+ MAU, microservices, event-driven async (RabbitMQ), cloud-native architecture, 120+ Agile sprints, mentored 5+ engineers. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Agentic AI & Generative AI: LangGraph, LangChain, Agentic AI Frameworks, Multi-Agent Orchestration, RAG Pipelines, Retrieval-Augmented Generation, LLM Integration, Prompt Engineering, Responsible AI, HITL (Human-in-the-Loop), Model Evaluation, LLM-as-Judge, Observability for AI, Agent Tracing, Hallucination Detection, OWASP LLM Top 10, Guardrails, Output Validation, Structured Outputs, Semantic Search, Hybrid Search, Vector Databases, ChromaDB, FastAPI, OpenAI API, Anthropic API, Groq, Ollama

Cloud, Backend & DevOps: AWS (S3, EC2, RDS, SQS, Lambda, API Gateway, Glue, Bedrock, SageMaker), Docker, Kubernetes (K8s), CI/CD (GitHub Actions), Terraform, Microservices, Cloud-Native Architecture, Event-Driven Architecture, Serverless, Application Security, Test Automation, Regression Harnesses, DevOps, Observability (Prometheus, Grafana, OpenTelemetry), RabbitMQ, Celery, Redis

Languages & Frameworks: Python (expert), Java, Spring Boot, PHP, TypeScript, JavaScript, SQL (expert), PL/SQL, R, Bash; Django, React.js, Node.js, GraphQL; PostgreSQL, MySQL, Oracle, SQLite, Snowflake; Git, Agile/Scrum, Jira

ML & Data: PyTorch, scikit-learn, XGBoost, Hugging Face, NLP, BERT, Anomaly Detection, Isolation Forest, Feature Engineering, MLflow, Tableau, Power BI

WORK EXPERIENCE

Blue Horse Digital
Sep 2026 – Present

AI/ML Engineer (Volunteer) — Agentic AI · ML Pipelines · Application Integration

- Developing agentic AI and ML pipelines for software applications - from data collection and feature engineering through model training, evaluation, and integration with back-end APIs and reporting layers.

Magizhs Tech Solution
May 2025 – Aug 2025

AI/ML Engineer Intern — Agentic AI · LangGraph · RAG · Responsible AI · Serverless

- Built an agentic mission-planning system as a **LangGraph stateful graph** (parse → plan → retrieve → validate → conditional routing) with a **RAG layer** over FAA Part 107 corpus - every compliance check functions as a hypothesis test grounded in cited regulation; **responsible AI guardrails** blocked non-compliant missions before MAVLink export (FAA altitude limits, battery budgets, geometric no-fly-zone enforcement).
- Parsed natural-language requests into structured flight plans via **serverless/event-driven** pipeline patterns; implemented **test automation** for compliance rule verification across mission scenarios.

George Mason University
Aug 2024 – Aug 2025

Research Associate — ML Pipelines · Anomaly Detection · Analytics Engineering

- Built multi-source prediction models (Python, R, 190+ countries) achieving **$R^2 = 0.977$**; applied **Isolation Forest anomaly detection** across 3 research pipelines; fine-tuned classification model (HuggingFace Transformers, PyTorch), deployed inference API on Quart, presented evaluation results to faculty. Designed Tableau/ggplot2 observability dashboards improving research comprehension 40%.

Entrata — Enterprise PropTech SaaS (AWS, 100K+ MAU)
Sep 2018 – Aug 2024

Senior Software Engineer — Enterprise Applications · AWS Cloud-Native · Microservices · CI/CD · DevOps

- Owned **enterprise production applications** powering 100K+ MAU on AWS - **cloud-native architecture**, **microservices** (Docker redesign: 50% scalability improvement, 30% deployment reduction), **event-driven async** via RabbitMQ with idempotency guarantees, and AWS S3/EC2/RDS/SQS across 120+ Agile sprints. Deployed Python + LLM **generative AI pipeline** on 1.6M regulated financial records with output validation gates; zero compliance incidents.
- Achieved **100% PHI compliance** on AWS S3 migration (Libsodium-encrypted pipelines, zero security incidents); **application security** across all resident data workflows; optimized 200+ SQL queries (30% performance gain); **test automation** via Python/Bash pipelines eliminating 80% manual operational workload.
- **Responsible AI + secure development** on regulated financial data: deterministic validation gates block silent AI failures before production; full audit logging; **CI/CD DevOps** with automated health checks. Mentored 5+ engineers; 300+ peer code reviews.

Entrata India
Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Full-Stack · APIs · AWS S3 · Enterprise Applications

- Built full-stack enterprise application modules (PHP, PostgreSQL, JavaScript, Java); designed SQL import pipelines for 1.6M HUD/Tax Credit records; AWS S3 document storage integration; compliance and certification workflows across 56 state-specific configurations.

PROJECTS

Leasel: Agentic AI Platform — 8-Agent LangGraph, RAG, HITL, Responsible AI
AGENTIC AI · RESPONSIBLE AI

Python · LangGraph · RAG · ChromaDB · FastAPI · OCR · CI/CD · LLM-as-Judge — github.com/venkatalakshmiparimala/Lease_complaince_rag

- **8-agent LangGraph pipeline** with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) - **responsible AI design**: LLM reasons; deterministic logic + **human-in-the-loop** gates make compliance calls; every answer cited to exact source clause or flagged unsupported. **93.6% precision, 87.9% F1** on 1,240 lease PDFs (5 US states); CI/CD regression harness blocks regressions before production.
- **RAG layer**: hybrid search (TF-IDF + semantic reranking), section-aware chunking, citation grounding, missing/conflicting clause detection - **production AI observability** via structured agent tracing across all 8 LangGraph nodes.

AgentProbe: AI Evaluation Toolkit — Model Evaluation, Observability Dashboard, CI/CD LLM EVALUATION LLMOPS

Python · LangGraph · FastAPI · Docker · GitHub Actions · React · ChromaDB · OpenAI · Anthropic · Groq · Ollama — github.com/venkatalakshmiparimala/AgentProbe

- Fully local, zero-vendor AI evaluation toolkit: **5-metric model evaluation engine** (correctness, groundedness, latency, token efficiency, hallucination rate), **CI/CD regression harness** with golden datasets blocking deploys on regressions, **observability dashboard** (React) tracking per-run metrics, retrieval drift, and latency across sessions - used to evaluate LeaseI across 1,240+ compliance test cases.
- **Multi-provider LLM integration** (OpenAI, Anthropic, Groq, Ollama) via 3-agent LangGraph runtime with tool calling and inter-agent state handoffs; **15-endpoint FastAPI backend**; Dockerized for consistent deployment; **prompt engineering** + LLM-as-judge calibration for eval reliability.

Magizhs Agentic Mission Planner — Serverless, Event-Driven, LangGraph, Responsible AI AGENTIC WORKFLOWS

Python · LangGraph · RAG · FastAPI · Serverless/Event-Driven Patterns · Test Automation

- Stateful LangGraph agentic system translating natural-language survey requests into structured flight plans via parse → plan → retrieve → validate → conditional routing; **RAG over regulatory corpus** (FAA Part 107) with cited compliance checks; **responsible AI guardrails** enforcing deterministic safety rules (altitude, battery, no-fly zones) before MAVLink export; serverless/event-driven dispatch patterns.

EDUCATION

M.S. Data Analytics Engineering — George Mason University — May 2026 · GPA 3.87

Agentic AI systems, ML pipelines, LLM evaluation, cloud-native architecture, data governance, statistical modeling

B.Tech Computer Science & Engineering — JNTU Anantapur — 2013 – 2017